

Quine McClusky Method - Tabulation Method

- Solved Problem #02

Simplify the following Boolean function by using a Quine McClusky method

$$F(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

Step 1: List all minterms in binary form as Column (a)

Step 2: Arrange the minterms according to number of 1's as Column (b)

Step 3: Compare each binary number with every term in the adjacent next higher Category and if they differ only by one position, put a check mark and copy the term in the next Column with '-' in the position that they differed

(a)		(b)		(c)	
Minterm	Binary	Minterm	Binary	Minterm	Binary
0	0 0 0 0	0	0 0 0 0 ✓	0, 1	0 0 0 - ✓
1	0 0 0 1	1	0 0 0 1 ✓	0, 2	0 0 - 0 ✓
2	0 0 1 0	2	0 0 1 0 ✓	0, 4	0 - 0 0 ✓
4	0 1 0 0	4	0 1 0 0 ✓	0, 8	- 0 0 0 ✓
5	0 1 0 1	8	1 0 0 0 ✓	1, 5	0 - 0 1 ✓
6	0 1 1 0	5	0 1 0 1 ✓	1, 9	- 0 0 1 ✓
8	1 0 0 0	6	0 1 1 0 ✓	2, 6	0 - 1 0 ✓
9	1 0 0 1	9	1 0 0 1 ✓	4, 5	0 1 0 - ✓
12	1 1 0 0	12	1 1 0 0 ✓	4, 6	0 1 - 0 ✓
13	1 1 0 1	13	1 1 0 1 ✓	4, 12	- 1 0 0 ✓
14	1 1 1 0	14	1 1 1 0 ✓	8, 9	1 0 0 - ✓
				8, 12	1 - 0 0 ✓
				5, 13	- 1 0 1 ✓
				6, 14	- 1 1 0 ✓
				9, 13	1 - 0 1 ✓
				12, 13	1 1 0 - ✓
				12, 14	1 1 - 0 ✓

Step 4: Apply the same process described in Step 3 for the resultant Column and Continue these cycles until a single pass through Cycle yields no further elimination of literals

Minterms

(d)

(e)

Minterm	Binary	Minterm	Binary
0, 1, 4, 5	0 - 0 - ✓	0, 1, 4, 5, 8, 9, 12, 13	- - 0 -
0, 1, 8, 9	- 0 0 - ✓		
0, 2, 4, 6	0 - - 0		
0, 4, 8, 12	- - 0 0 ✓		
1, 5, 9, 13	- - 0 1 ✓		
4, 5, 12, 13	- 1 0 - ✓		
4, 6, 12, 14	- 1 - 0		
8, 12, 9, 13	1 - 0 - ✓		

Step 5: List the prime implicants

Prime Implicants		Binary			
$\bar{A}\bar{D}$	0, 2, 4, 6	0	-	-	0
$B\bar{D}$	4, 6, 12, 13	-	1	-	0
$\bar{C}$	0, 1, 4, 5, 8, 9, 12, 13	-	-	0	-

Step 6: Select the minimum number of prime implicants which must cover all the minterms

Table shows prime implicant selection chart. Each prime implicant is represented in a row and each Column represented by minterm. Dots are placed in each row to show the composition of minterms that makes the prime implicants. From this chart we have to select the minimum number of prime implicants which must cover all minterms. The selection procedure is as follows.

Search for single dot Column and select the prime implicants corresponding to that dot by putting the check mark in front of it.

Prime Implicants			m0	m1	m2	m4	m5	m6	m8	m9	m12	m13	m14
			c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11
$\overline{A}\overline{D}$	0, 2, 4, 6	✓	⊙		⊙	⊙		⊙					
$B\overline{D}$	4, 6, 12, 14	✓				⊙		⊙			⊙		⊙
$\overline{C}$	0, 1, 4, 5, 8, 9, 12, 13	✓	⊙	⊙		⊙	⊙		⊙	⊙	⊙	⊙	
			x			x		x			x		

Search for multiple dot columns one by one.
If the corresponding minterms is already included in the final expression ignore the minterm and go to next multi dot column, otherwise include the corresponding prime implicant in the final expression

Column c1(m0) has multiple dots, so ignore

Column c2(m1) has single dot, so include corresponding prime implicant (0, 1, 4, 5, 8, 9, 12, 13)

Column c3(m2) has single dot, so include corresponding prime implicant (0, 2, 4, 6)

Column c4(m4) has multiple dots, so ignore

Column c5(m5) has single dot, but the corresponding minterm is already included in the final expression

Column c6(m6) has multiple dots, so ignore

Column c7(m8) and c8(m9) has single dot, but the corresponding minterm is already included in the final expression

Column c9(m12) has multiple dots, so ignore

Column 10 (m_{13}) has single dot, but corresponding minterm is already included in the final expression

Column 11 (m_{14}) has single dot, so include corresponding prime implicant - (4, 6, 12, 14)

$$F(A, B, C, D) = 0-0-0 + -1-0 + --0-$$

$$F(A, B, C, D) = \bar{A}\bar{D} + B\bar{D} + \bar{C}$$